# 1. Problem, motivation and project team

## Kidney stone segmentation

- Delineate annotated stone pixels in axial CT exports.
- Pixel masks retain location and extent for image analysis.
- Manual outlining motivates automatic segmentation (Bouzon et al., 2025).
- Evaluate reproducibility and a focused attention modification.

No clinical deployment or patient diagnosis claim.

**22AIE437: Medical Image Processing**

Amrita School of Artificial Intelligence

Group: [REQUIRED]
Names and roll numbers: [ALL REQUIRED]

Submission draft; team details pending.

# 2. Dataset, objectives and evaluation

- KSSD2025: 838 matched TIFF pairs; 831 eligible after seven duplicate exclusions.
- 512×512 grayscale; stone vs background.
- Image /255; decoded mask $> 127$; no resizing.
- Five content-group outer folds; separate inner checkpoint selection.

**Frozen fold sizes**

| Fold | Train | Inner | Outer |
|---|---|---|---|
| 1 | 446 | 206 | 179 |
| 2 | 455 | 97 | 279 |
| 3 | 665 | 48 | 118 |
| 4 | 589 | 70 | 172 |
| 5 | 606 | 142 | 83 |

$D = 2TP/(2TP + FP + FN)$; $J = TP/(TP + FP + FN)$.
Precision and recall use the same prediction threshold $p \geq 0.5$.

Sources: frozen dataset manifest and splits. Groups are content clusters, not patient IDs. No additional external project test cohort.

# 3. Base paper and research context

**KSSD2025: A New Annotated Dataset for Automatic Kidney Stone Segmentation and Evaluation With Modified U-Net Based Deep Learning Models**

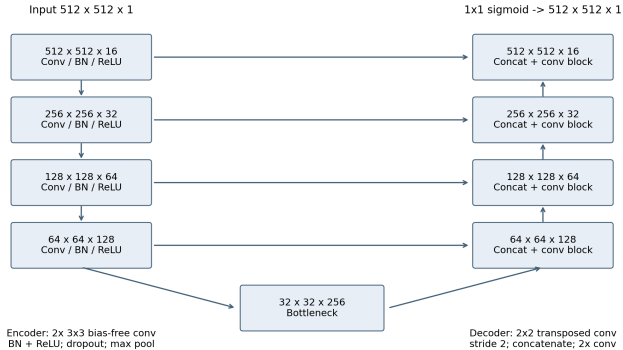
Murillo F. Bouzon; Paulo H. S. de Santana; Gabriel N. Missima; Weverson S. Pereira; Fernando P. Rivera; Gilson A. Giraldi; Oscar E. H. Fugita; Paulo S. S. Silva.

*IEEE Access*, volume 13, 171790–171806 (2025).
DOI: 10.1109/ACCESS.2025.3610027.

- Annotated 838 axial stone-containing CT images.
- Evaluated lightweight U-Net, U-Net++, U-Net3+ and TransU-Net.
- Reported strong overlap; also analyzed errors and external samples.

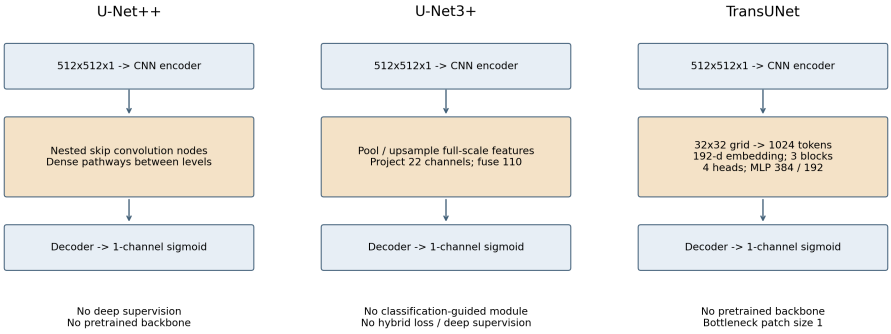
Primary source inspected directly: supplied paper, title p.1, methodology pp.3–8, Table 3 p.9. Paper and dataset contributions belong to their authors.

# 4. Replication baseline: lightweight Modified U-Net




Attributed redrawing of Bouzon et al. Fig. 5 with local code-specific blocks. Encoder 16/32/64/128; bottleneck 256. Two bias-free 3x3 convolutions, BN and ReLU; encoder dropout .1/.2/.3/.3, bottleneck .3.

# 5. Key blocks and three implemented alternatives




Baseline decoder: 2x2 transposed convolution, skip concatenation, conv block, 1x1 sigmoid. Library U-Net++ and U-Net3+ use interpolated upsampling. All output 512x512x1 probabilities.

Local implementations inspired by Zhou et al., Huang et al. and Chen et al. Deep supervision and pretrained backbones are disabled. Source: kssd/models.py.

# 6. Literature versus our measured overlap results

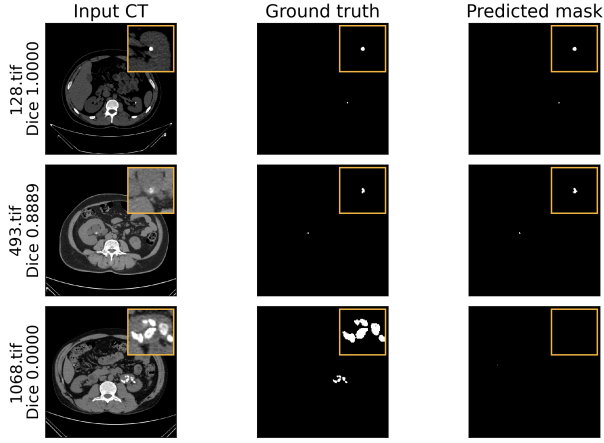
Table: Mean Dice and IoU: literature and local corrected evaluation.

| | Paper Dice | Our Dice | Paper IoU | Our IoU |
|---|---|---|---|---|
| Modified U-Net | .9706 | .7910 | .9465 | .6589 |
| U-Net++ | .9663 | .3678 | .9397 | .2490 |
| U-Net3+ | .9668 | .3230 | .9402 | .2805 |
| TransUNet | .9553 | .4842 | .9221 | .3749 |

- Baseline has the highest measured mean among our four models.
- All folds, including failures, remain in the averages.
- These results do not establish equivalence to the literature protocol.

Paper Table 3 (primary PDF p.9), canonical summary.json files. Our mean is unweighted over five fold-pooled binary scores. Full SDs and protocol details are in the report; numerical superiority is not statistically established.

# 7. Three verified Input / Ground Truth / Prediction examples




Attention model, original saved outer predictions: 128.tif success, 493.tif middle-ranked Dice, 1068.tif largest FN. Crops supplement the three required full views. Independent ground-truth intersections verified.

# 8. Working code and traceable experiment records

- Data: data.py / validate_data.py
- Model: models.py
- Training/evaluation: train.py
- Metrics: metrics.py / tf_metrics.py
- Records: reporting.py / auditors
- New seeded model and optimizer per fold.
- Earliest inner hard-Dice maximum selects weights.

**Baseline fold 1: original log**

| Epoch | Loss | Inner Dice |
|---|---|---|
| 1 | 0.9916 | 0.0000 |
| 5 | 0.1852 | 0.7207 |
| 9 | 0.1486 | 0.8825 |
| 10 | 0.1156 | 0.7800 |

Selected epoch: 9 (inner maximum)

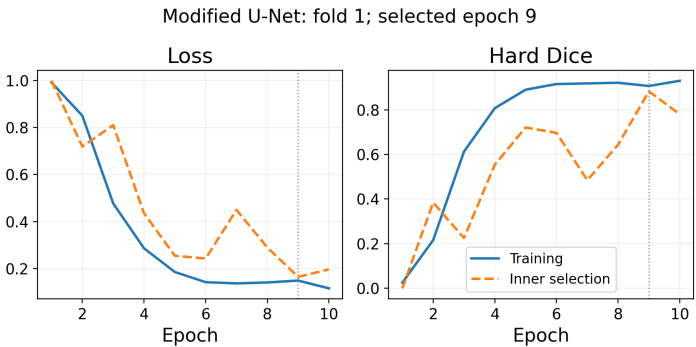
Original fold-1 CSV excerpt, selected rows. All ten epochs are in the report.

Demo: `python -m kssd.attention_audit --output runs/demo.json`
GitHub implementation and compact evidence. New inference needs official data and recovered local selected weights.

# 9. Training settings and loss/metric curves

TensorFlow 2.20 / Keras 3.13.2
RTX 4070 Laptop, WSL
Adam; LR 0.001; float32
Batch size 2; Dice loss
Base seed 42; fold seeds 43–47
Train-only paired augmentation

**250 completed training epochs across 25 fold runs.**
Each fold run: 10 epochs.
Five folds per model, five models.




Training soft loss and hard binary metrics are distinct. Curves use training and inner-selection data; outer test does not choose the epoch. Full five-model curves and exact settings are in the report.

# 10. Our architectural modification

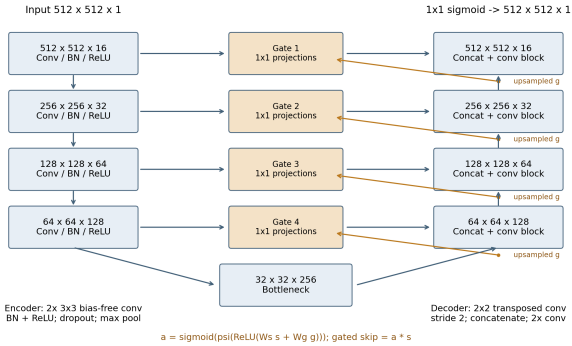
**Four additive attention gates on baseline skip connections.**

- Research motivation: suppress irrelevant bright structures while preserving foreground detail.
- Adapted from Oktay et al., *Attention U-Net* (2018).
- Gate each encoder skip using upsampled decoder context.
- Preserve encoder/decoder block design, widths, dropout and normalization.
- Architecture is the only changed configuration field.

**Contribution boundary**

The team study applies a published mechanism to this baseline and evaluates it fairly. It does not invent attention, KSSD2025 or the paper's Modified U-Net.

# 11. Modified architecture and gate equation




$$a = \sigma(\psi[\mathrm{ReLU}(W_s * s + W_g * g)]), \quad \tilde{s} = a \odot s.$$

Baseline: 1,945,233 parameters; Attention: 1,967,357 (+22,124).

Identical splits, seeds, augmentation, loss, optimizer, budget and checkpoint policy.

# 12. Controlled comparison: an honest negative result

Table: Fold mean and sample SD; no significance claim.

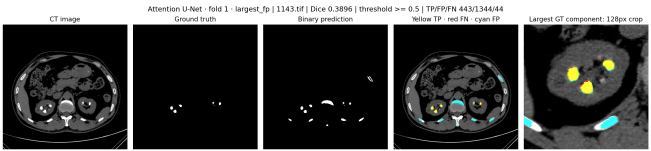
| Model | Dice | IoU |
|---|---|---|
| Modified U-Net | $.7910 \pm .0724$ | $.6589 \pm .0973$ |
| Attention U-Net | $.7585 \pm .2205$ | $.6515 \pm .2864$ |

- Mean Dice change: $-0.032463$ (Attention minus baseline).
- All five folds retained; fold differences vary in sign.
- Negative mean result: no improvement or superiority claim.
- Baseline/no gates versus all gates is a controlled comparison; gate-level contributions are not isolated.

Guidelines accept a reasoned negative result. Improvement ablation is conditional; no improvement is claimed here.

Source: comparison.json and independently checked per-fold records.

# 13. Failure analysis and next experiments

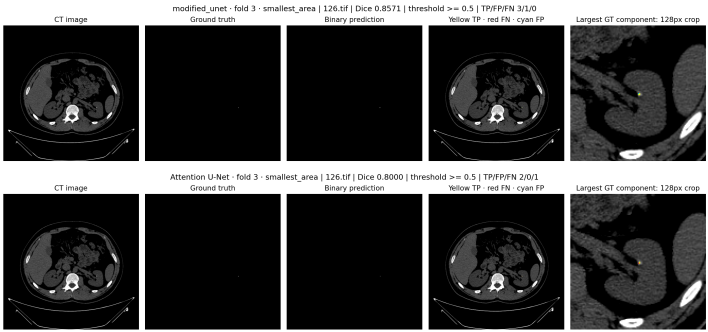



1143.tif: TP443 / FP1344 / FN44; Dice .3896.

- False positives on bright background structures; complete misses also occur.
- Tiny masks and limited optimization may contribute.
- Gating/normalization dynamics remain unproven causes.
- Next: prespecified longer budget, multiple seeds, gate placement tests.
- Focal Tversky: zero completed folds; performance not measured.

Do not tune using observed outer scores or discard bad folds. Recorded failures support descriptions; they do not identify a causal mechanism.

# 14. Matched small-stone case: one pixel matters




**126.tif: 3 foreground pixels**

Baseline: TP/FP/FN 3/1/0
Dice .8571; IoU .75

Attention: 2/0/1
Dice .8; IoU .6667

Pixel area is not physical size. One illustrative case does not establish size-group superiority.

Same outer fold and sample. Selected-checkpoint panels and exact per-image counts from the baseline and Attention figure indexes.

# 15. Conclusions, contributions and references

- Modified U-Net leads our four-model mean Dice.
- Skip attention lowers mean overlap under the matched budget.
- 250 training epochs across 25 completed runs; preserve negative evidence.
- Limits: no patient IDs/external cohort; tiny masks; weights local.
- Next: team details, recoverable weight distribution, longer controlled study.

**Contributions: [EACH MEMBER / ACTUAL ROLE REQUIRED].**
Every member presents and prepares for all questions.

**References**


Bouzon et al. IEEE Access 13, 2025. DOI 10.1109/ACCESS.2025.3610027.

Oktay et al. Attention U-Net, 2018. arXiv:1804.03999.

Zhou et al. UNet++, 2018. arXiv:1807.10165.

Huang et al. UNet 3+, 2020. arXiv:2004.08790.

Chen et al. TransUNet, 2021. arXiv:2102.04306.


Full verified BibTeX in references.bib; complete discussion in report.